

## RESEARCH

## A Map of 75 Human Ribosomal Protein Genes

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We mapped 75 genes that collectively encode >90% of the proteins found in human ribosomes. Because localization of ribosomal protein genes (rp genes) is complicated by the existence of processed pseudogenes, multiple strategies were devised to identify PCR-detectable sequence-tagged sites (STSs) at introns. In some cases we exploited specific, pre-existing information about the intron/exon structure of a given human rp gene or its homolog in another vertebrate. When such information was unavailable, selection of PCR primer pairs was guided by general insights gleaned from analysis of all mammalian rp genes whose intron/exon structures have been published. For many genes, PCR amplification of introns was facilitated by use of YAC pool DNAs rather than total human genomic DNA as templates. We then assigned the rp gene STSs to individual human chromosomes by typing human-rodent hybrid cell lines. The genes were placed more precisely on the physical map of the human genome by typing of radiation hybrids or screening YAC libraries. Fifty-one previously unmapped rp genes were localized, and 24 previously reported rp gene localizations were confirmed, refined, or corrected. Though functionally related and coordinately expressed, the 75 mapped genes are widely dispersed: Both sex chromosomes and at least 20 of the 22 autosomes carry one or more rp genes. Chromosome 19, known to have a high gene density, contains an unusually large number of rp genes (12). This map provides a foundation for the study of the possible roles of ribosomal protein deficiencies in chromosomal and Mendelian disorders.

[The sequence data described in this paper have been submitted to GenBank. They are listed in Table 1.]

Although the ribosome, as catalyst for protein synthesis, is known to be essential for organismal growth and development, the effects of ribosomal mutations and their role in human disease have been explored barely. The mammalian ribosome is a massive structure composed of 4 RNA species and ~80 different proteins (Wool 1979). Typical mammalian cells contain about  $4 \times 10^6$  ribosomes, and ribosomal RNAs and proteins constitute ~80% of all cellular RNA and 5%–10% of cellular protein. One might predict that genetic defects in ribosomal components would invariably result in early embryonic death. However, there is strong evidence in *Drosophila* and suggestive evidence in humans that quantitative deficiencies of ribosomal proteins may yield viable but abnormal phenotypes. In *Drosophila*, the *Minute* phenotype (reduced body size,

diminished fertility, and short, thin bristles) results from heterozygous deficiencies (deletions) at any 1 of 50 loci scattered about the genome (Schultz 1929; FlyBase 1997). Several *Minute* loci have been characterized molecularly, and all have been found to encode ribosomal proteins (Kongsuwan et al. 1985; Hart et al. 1993; Andersson et al. 1994; Cramton and Laski 1994; Saebøe-Larsen and Lambertsson 1996; Schmidt et al. 1996; A. Cheng, A. Zinn, J. Mach, R. Lehman, and D.C. Page, unpubl.). Thus, it appears that reductions in the amount of any of a number of ribosomal proteins have a similar, characteristic effect on the development of *Drosophila* embryos.

Perhaps ribosomal protein deficiencies have analogous consequences in humans, resulting in specific, recognizable clinical features (which might or might not resemble the *Minute* phenotype observed in *Drosophila*). We and our colleagues have reported findings consistent with a role for ribosomal protein S4 (RPS4) deficiency in the etiology of

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certain features of Turner syndrome, a complex human disorder classically associated with a 45,X karyotype (Fisher et al. 1990; Watanabe et al. 1993; Zinn et al. 1994). We are intrigued by the possibility that deficiencies of other human ribosomal protein genes (rp genes) might cause phenotypic abnormalities similar to those seen in Turner syndrome—just as deficiencies of any of a number of *Drosophila* rp genes cause the *Minute* phenotype.

The ribosome is the largest, most complex mammalian structure to be completely described at the level of nucleotide and amino acid sequence. The nucleotide sequences of the four ribosomal RNAs—28S, 18S, 5.8S, and 5S—have been determined in their entirety (Maidak et al. 1997), and a systematic effort to deduce the primary structure of all mammalian ribosomal proteins by cDNA sequencing has come to completion (Wool et al. 1996).

Moreover, the genes encoding the RNA constituents of the mammalian ribosomes have all been assigned to chromosomes. The 28S, 18S, and 5.8S rRNAs are generated by elaborate processing of a single 45S precursor derived from tandemly repeated gene arrays which, in humans, are located on the short arms of chromosomes 13, 14, 15, 21, and 22 (Henderson et al. 1973; Worton et al. 1988). The 5S rRNA derives from tandemly repeated gene clusters on human chromosome 1 (Sørensen et al. 1991; Lomholt et al. 1995).

Paradoxically, only a small fraction of the genes encoding the mammalian ribosomal proteins have been mapped previously. Though these 80 proteins function together, their amino acid sequences are dissimilar. Unlike the ribosomal RNAs, each mammalian ribosomal protein typically is encoded by a single gene. However, in the case of most if not all ribosomal proteins, the single, functional gene has generated a large number of silent, processed pseudogenes at sites dispersed throughout the genome (Dudov and Perry 1984; Wagner and Perry 1985; Kuzumaki et al. 1987). These pseudogenes impede the mapping of the functional rp genes, explaining at least in part, why only 24 of the ~80 rp genes had been chromosomally assigned. The 24 genes that had been assigned map to 14 different chromosomes, suggesting that rp genes, unlike rRNA genes, are not clustered at a few sites in the genome (Feo et al. 1992).

If we are to explore systematically the possibility that ribosomal protein deficiencies or mutations cause certain human disorders, we must first learn the chromosomal map position of each of the ~80 human rp genes. This task is hindered by the exist-

ence of processed pseudogenes elsewhere in the genome. We developed general strategies to physically map human rp genes, while avoiding pseudogenes, using sequence tags specific to the functional, intron-bearing genes.

## RESULTS

The human rp genes had been previously characterized to varying degrees. Some human rp genes had been completely sequenced at both the cDNA and genomic levels, whereas for others, even partial cDNA sequences were unavailable. We divided the estimated 80 human rp genes into three classes, ordered according to how much gene structure information was available (Table 1). For group 1, containing 19 genes, at least some human splice sites had been determined and some human intron sequence was available. No human splice site information was available for any of the remaining 61 rp genes. However, for 12 of these 61 genes, at least some splice sites had been determined in the homologous genes in rat, mouse, chicken, or frog. As described below, the extreme conservation of splice-site positions among homologous vertebrate rp genes allowed us to predict the positions of splice sites in the human genes using this information. These 12 rp genes, for which somewhat less information was available, constitute group 2. No vertebrate splice-site information was available for any of the remaining 49 rp genes, which comprise group 3.

For each of the three groups, we developed a separate strategy for generating sequence tags specific to the functional, intron-bearing genes. For all three groups, we exploited the fact that rp pseudogenes, derived from processed transcripts, lack the introns found in their progenitors (Dudov and Perry 1984; Davies et al. 1989). For group 1 genes, identification of STSs was straightforward. We derived STSs specific to the functional genes by choosing oligonucleotides from the previously sequenced introns (Fig. 1A).

### STSs for Group 2 and 3 Genes: Intron Trapping

For group 2 and group 3 genes, no human intron sequences were available. However, other information allowed us to predict the location of, and then trap, introns from these genes. The methods we employed were inspired in part by earlier successes in identifying and mapping intron-bearing rp genes by PCR (Davies et al. 1989).

For group 2 genes, the positions of at least some splice sites in vertebrate homologs were known, and this information played a central role in our map-

## A MAP OF 75 HUMAN RIBOSOMAL PROTEIN GENES

Table 1. Chromosomal Assignments and STSs for Human Ribosomal Protein Genes

| Gene   | Chrom-<br>osome | Group | Forward primer <sup>a</sup> | Reverse primer <sup>a</sup> | PCR prod.<br>size (bp) | STS name  | Accession no. <sup>b</sup>                    |
|--------|-----------------|-------|-----------------------------|-----------------------------|------------------------|-----------|-----------------------------------------------|
| RPSA   | 3               | 3     | GGAGGAATTTACGGTGAAT         | AGACAGTCAGTGGTTCCTC         | 426                    |           | J03799                                        |
|        |                 |       | GGAGGAATTTACGGTGAAT         | ccactcccaacagcagtccta       | 272                    | RP_SA_1   | AB007146 <sup>d</sup>                         |
| RPS2   | 16              | 3     | CTGCTCATGATGCGCTGGT         | CACAGCTGGAGCCCTGAGT         | 460                    |           | X17206                                        |
|        |                 |       | AACTTCGgttaggtgggtccac      | caggaggggtcagtggtgtg        | 214                    | RP_S2_1   | AB007147 <sup>d</sup>                         |
| RPS3   | 11 <sup>e</sup> | 1     | catgtgggtcccaoctattcc       | gggggaaaagtgcacattca        | 263                    | RP_S3     | L16016                                        |
| RPS3A  | 4 <sup>i</sup>  | 3     | CCCTCCATGATGCTCTGGT         | TTCTTGGACTGGTGGTTCAT        | 358                    |           | M77234                                        |
|        |                 |       | aaatcacatgatttctgttaggg     | CGTTTCCCActacaaggcaa        | 187                    | RP_S3A_1  | AB007148 <sup>d</sup>                         |
| RPS4X  | X <sup>e</sup>  | 1     | tgagatggattgaatgtggc        | ttaaagaggggtgcccaggtta      | 249                    | RP_S4X    | AF041428                                      |
| RPS4Y  | Y <sup>e</sup>  | 1     | tttaaggggacagatttcaacttc    | ccacatttaaaactgagtagcgtcc   | 361                    | RP_S4Y    | AF041427                                      |
| RPS5   | 19              | 3     | CATCTGGCTGCTGTGCAC          | GTITGGACTTGGCCACACGCT       | 246                    |           | U14970                                        |
|        |                 |       | CTCATCAATGCTGCCAAGgt        | ccacaaatgcccacttaag         | 66                     | RP_S5     | AB007149 <sup>d</sup>                         |
| RPS6   | 9 <sup>h</sup>  | 1     | aaagccattggtaactggta        | agctgcattttctaacagatg       | 325                    | RP_S6     | X67309                                        |
| RPS7   | 2 <sup>i</sup>  | 1     | agggttcagccacagtgagag       | acccacccctaactaccag         | 249                    | RP_S7     | Z25749                                        |
| RPS8   | 1 <sup>i</sup>  | 1     | tctcagttcctttgactgcc        | atctctgtgggttcagggtacc      | 307                    | RP_S8     | X67247                                        |
| RPS9   | 19              | 3     | TGCCAAACTTATGTGACCC         | CTTCGAACAGACGCCGTG          | 399                    |           | U14971                                        |
|        |                 |       | agtcogtccctaaatttgg         | CTTCGAACAGACGCCGTG          | 245                    | RP_S9     | AB007150 <sup>d</sup>                         |
|        |                 |       | COGGAATGCCATCTATGAC         | GCAGATGAAGGTATCAAGG         | 468                    |           | U14972                                        |
|        |                 |       | atggagggtatcttccoctaa       | GCAGATGAAGGTATCAAGG         | 266                    | RP_S10_2  | AB007151 <sup>d</sup>                         |
| RPS10  | 6               | 3     | CTATCTGCACTACATCCGCA        | CTTCTGGACTGCTTCTTGG         | 1663                   |           | X06617                                        |
| RPS11  | 19 <sup>k</sup> | 3     | ttccagatcggaaccaattta       | tactgcgaatttcaactgt         | 1287                   | RP_S11    | AB007152 <sup>d</sup>                         |
| RPS12  | 6               | 3     | CTTTTGGCTGAACACCAAT         | TCTTCATGACATCCTTGGC         | 1100 <sup>c</sup>      |           | X53505                                        |
|        |                 |       | gaagtcgttaaggtccctgag       | TCCTCAATGACATCCTTGGC        | 206                    | RP_S12    | AB007153 <sup>d</sup>                         |
| RPS13  | 11              | 3     | ATGCTAAATTCGGTCTGATTC       | GGCAGAGGCTGTAGATGATT        | 745                    | RP_S13    | L01124                                        |
| RPS14  | 5 <sup>j</sup>  | 1     | ttccatttagcctgccttttg       | ccccagaaactaccacagc         | 388                    | RP_S14_1  | ML3934                                        |
| RPS15  | 19              | 1     | agacaggtttcagtggtgcc        | agaccacagcctcagacaag        | 319                    | RP_S15    | M32405                                        |
| RPS15A | 16              | 3     | CATGAATGTCTGGCAGATG         | CATCAATGATTTCAAATTCGC       | 900 <sup>c</sup>       |           | X84407                                        |
|        |                 |       | ttgttttgggtattgtctccc       | CATCAATGATTTCAAATTCGC       | 236                    | RP_S15A_1 | AB007154 <sup>d</sup>                         |
| RPS16  | 19              | 2     | GGTGGCTATATACCAAAAGgt       | GGAGCCCTCATCCACATct         | 130 <sup>c</sup>       | RP_S16_1  | ML1408, M60854                                |
| RPS17  | 15 <sup>z</sup> | 1     | tcacccagctagtaagtggc        | gtgggttagcatggagacacc       | 281                    | RP_S17    | ML8000                                        |
| RPS18  | 6               | 2     | GATGGAAATACAGCCAGgt         | GCCTCGGACACGAGGct           | 350 <sup>c</sup>       | RP_S18    | M76762, X69150                                |
| RPS19  | 19              | 3     | AGATCTATGGGGAGGTGAG         | CTGTCCGGGATTCCTGT           | 650 <sup>c</sup>       |           | M81757                                        |
|        |                 |       | acctgactagggccctcagt        | CTGTCCGGGATTCCTGT           | 191                    | RP_S19_1  | AB007155 <sup>d</sup>                         |
| RPS20  | 8               | 3     | TGAAAGGACCCAGTTCGAATG       | TCAACTCTCTGGCTCAACTG        | 607                    |           | L06498                                        |
|        |                 |       | TGAAAGGACCCAGTTCGAATG       | caaccaagaacacagcaacaa       | 233                    | RP_S20    | AB007156 <sup>d</sup>                         |
| RPS21  | 20              | 3     | CAATGCGATCATCGGTGC          | CTITGGCCATCGGAGAAT          | 723                    |           | L04483                                        |
|        |                 |       | gggtgaaggtacagggcagagg      | gatgtcccggaagtgtatgagc      | 199                    | RP_S21    | AB007157 <sup>d</sup>                         |
| RPS22  | 5               | 3     | CTTTGTACCAATGACGGTT         | ATCTTGTGCTTTCTCTTGC         | 324                    |           | D14530                                        |
|        |                 |       | ttctgttgggtgggtggc          | ttatttctgggtgttgggtggc      | 101                    | RP_S23_1  | AB007158 <sup>d</sup>                         |
| RPS24  | 10 <sup>m</sup> | 2     | CCCAACATGAGCTTGCAGAGgt      | CTTCTCATACAGGCCATGct        | 1300 <sup>c</sup>      |           | X71972, M31520                                |
|        |                 |       | gggaaagaccaaagcaatctg       | ccaggtctacctttccaaaca       | 217                    | RP_S24_2  | AB007159 <sup>d</sup>                         |
| RPS25  | 11 <sup>n</sup> | 1     | AGGAGCTCTCTTAAAGgtgag       | TGAAACCCAGCTTGAATGct        | 1200 <sup>c</sup>      | RP_S25    | D15048, M64716                                |
| RPS26  | 12              | 3     | AAATTCGTCAATTCGAACAT        | AGCAACCCAGGCTTAAATC         | 900 <sup>c</sup>       |           | X69654                                        |
|        |                 |       | AAGCGAGCGTCTTCGATGgt        | GCTTGGGAGGCATAGGct          | 750 <sup>c</sup>       | RP_S26    | AB007160 <sup>d</sup> , AB007161 <sup>d</sup> |
| RPS27  | 1               | 3     | GATCTCTCTATCCCTCTCC         | GTAAGCCTTGTCTTTCTCTC        | 550                    |           | L19739                                        |
|        |                 |       | cgtatccttgaagctgtgca        | gtttcccaacacaaaaaacg        | 184                    | RP_S27_2  | AB007162 <sup>d</sup>                         |
| RPS27A | 2               | 3     | GTTAAGCTGGCTGTCTGAA         | GTAAGTCAGACAACATTTGCCA      | 618                    |           | X63237                                        |
|        |                 |       | GTTAAGCTGGCTGTCTGAA         | tttcacagaataacctcatttaaac   | 169                    | RP_S27A_1 | AB007163 <sup>d</sup>                         |
| RPS28  | 19              | 3     | CGAGTGTCTACCCCTTTTG         | GAGCATCTCAGTTAGGTGTGG       | 342                    |           | L05091                                        |
|        |                 |       | ttgatagaccttgggtgggtg       | acatgagatgttgacaggcc        | 141                    | RP_S28_1  | AB007164 <sup>d</sup>                         |
| RPS29  | 14              | 3     | ACGGTCTGATCCGAAATAT         | AAGAATTATCATGGTTTTCATTG     | 2300 <sup>c</sup>      |           | U14973                                        |
|        |                 |       | ACGGTCTGATCCGAAATAT         | aaagggttaggttaagactaagtc    | 310                    | RP_S29    | AB007165 <sup>d</sup>                         |
| RPS30  | 11 <sup>e</sup> | 1     | aagcgtctagtgtgtggg          | ggcctgtgtctgacctaatct       | 231                    | RP_S30    | X65921                                        |
| RPL3   | 22              | 3     | CCAATGACTTTGTCTATGCTG       | TTGCAATTCGGTCTTCTCTC        | 745                    |           | X73460                                        |
|        |                 |       | ggcaggtcctgacttgtcat        | TTGCAATTCGGTCTTCTCTC        | 673                    | RP_L3_1   | AB007166 <sup>d</sup>                         |
| RPL4   | 15              | 2     | GCTTCCCTCAAGAGTAACAAgt      | ATCTTGTGATGGGAGAGct         | 600 <sup>c</sup>       |           | M15678, D23660                                |
|        |                 |       | GCTTCCCTCAAGAGTAACAAgt      | gtggtggctcaccctataaa        | 400 <sup>c</sup>       | RP_L4_2   | AB007167 <sup>d</sup>                         |
| RPL5   | 1               | 2     | CAAACAGAGATTCATTTGTGAGgt    | TCATATACGGCATTAAGCAATct     | 1200 <sup>c</sup>      | RP_L5     | D10737, U14966                                |
| RPL6   | 12              | 3     | CCCAAAACATCTTACTGATGC       | GATAAATTCATTCGTGAGAGC       | 700                    | RP_L6     | D17554                                        |
| RPL7   | 8               | 1     | ggtcagtgcttgcagtagt         | ctcccgtagctgagcagtggt       | 296                    | RP_L7     | L16557                                        |
| RPL7A  | 9 <sup>p</sup>  | 1     | atgcttcttctgttcttattgg      | cactaggaaggtgtgtgtg         | 299                    | RP_L7A    | X52138                                        |
| RPL8   | 8               | 3     | TGATCCGTGGACAGAGGAAG        | TGCAATAGTTCCCTGATGC         | 747                    |           | Z28407                                        |
|        |                 |       | gaggaggttaagctcctaaacg      | gcagtatcaggccggcag          | 206                    | RP_L8_1   | AB007168 <sup>d</sup>                         |
| RPL9   | 4 <sup>q</sup>  | 3     | CATTTAGCTTTGTTCAAATTC       | TCATCTGCATCTTCTTTCTG        | 693                    |           | D14531                                        |
|        |                 |       | aatggagaggttaagcattctgg     | cttggccaggaaagtactaatc      | 170                    | RP_L9_1   | AB007169 <sup>d</sup>                         |

(Continued on following page.)

Table 1. (Continued)

| Gene   | Chromosome      | Group | Forward primer <sup>a</sup> | Reverse primer <sup>a</sup> | PCR prod. size (bp) | STS name  | Accession no. <sup>b</sup>                   |
|--------|-----------------|-------|-----------------------------|-----------------------------|---------------------|-----------|----------------------------------------------|
| RPL10  | X <sup>i</sup>  | 3     | TATCATGTGTCATCGCACC         | TGGACAGGAAGTAGAATTTATGG     | 379                 |           | M73791                                       |
|        |                 |       | TATCATGTGTCATCGCACC         | ggagcaatggaaagaactga        | 163                 | RP_L10    | AB007170 <sup>d</sup>                        |
| RPL10A | 6               | 1     | gcacogttctgatccac           | CCAGGACACACAGAGAA           | 395                 | RP_L10A   | U12404,Z62020                                |
| RPL11  | 1               | 3     | ATCCGCAAACTCTGTCTCAA        | GTCAGTGGACAGCAATCTT         | 1000 <sup>c</sup>   |           | L05092                                       |
|        |                 |       | ATCCGCAAACTCTGTCTCAA        | gactcgacagcaggttaaatca      | 187                 | RP_L11_1  | AB007171 <sup>d</sup>                        |
| RPL12  | 9               | 3     | ATGTGTAACATGTCTGAC          | GCATTCACAGCAOCCACT          | 509                 | RP_L12    | L06505                                       |
| RPL13  | 16 <sup>a</sup> | 3     | AACTGAAACTGGCCACC           | TTCAACATCCGTCTCTGG          | 689                 |           | X64707                                       |
|        |                 |       | CATGCCCGTCCGGAAGgt          | CGAGCTTTCTCTCTTATAGACcta    | 531                 | RP_L13    | AB007172 <sup>d</sup>                        |
| RPL13A | 19              | 2     | GCTAAACAGGTACTGCTGGgt       | AGGAAAGCCAGGTACTTCAct       | 592                 | RP_L13A   | X51528,X56932                                |
| RPL15  | 3               | 3     | TCTGATGTCAATGCGCTTTCT       | GTTAACAACATGATGGACAGG       | 600 <sup>c</sup>    |           | L25899                                       |
|        |                 |       | TCTGATGTCAATGCGCTTTCT       | agagagacactgagggcagca       | 223                 | RP_L15_1  | AB007173 <sup>d</sup>                        |
| RPL17  | 18              | 3     | CATGGTGGGTTAACCCATA         | CCGTGCCATAAGTTTGTGT         | 900 <sup>c</sup>    |           | X53777                                       |
|        |                 |       | ccccacttagatgtacatagacc     | tgaggagacttcagcttattctgt    | 235                 | RP_L17    | AB007174 <sup>d</sup>                        |
| RPL18  | 19              | 2     | CGCACGCCACACCAAggt          | CGACGCGAGGTAGGGTct          | 400 <sup>c</sup>    | RP_L18    | X05025,L11566                                |
| RPL18A | 19              | 3     | CCCACTCCATTCAGATCATG        | TTTATTTGGGGCACACC           | 348                 |           | L05093                                       |
|        |                 |       | ctccccctggaggggaagtgt       | aggcatgttgaggggggtt         | 113                 | RP_L18A   | AB007175 <sup>d</sup>                        |
| RPL19  | 17 <sup>i</sup> | 2     | AGATCGATGCGCACATgt          | TCAGGTACAGGCTGTGATACct      | 900 <sup>c</sup>    | RP_L19    | M30264,X63527                                |
| RPL21  | 13              | 3     | ACCTGGGTTCAACTAAGCG         | GAATAGGTTCCAGCAGCTCA        | 195                 |           | U14967                                       |
|        |                 |       | ACCTGGGTTCAACTAAGCG         | GGTGGAGCAGGctattaaagg       | 136                 | RP_L21_1  | AB007176 <sup>d</sup>                        |
| RPL23  | 17              | 3     | TGCTCATTCGACAAAGAA          | CAGCATTTGATGCAATCC          | 318                 | RP_L23    | X52839                                       |
| RPL23A | 17 <sup>u</sup> | 1     | atacattaccogccccctctc       | ggtgcacatcagtggttctctt      | 231                 | RP_L23A   | AF001689                                     |
| RPL24  | 3               | 3     | TGCTGCATCTCTTCTGTATA        | TTAGCGTTTTCACCAACTC         | 1600 <sup>c</sup>   |           | M94314                                       |
|        |                 |       | tctggatgtgagagatgggt        | TTAGCGTTTTCACCAACTC         | 155                 | RP_L24_1  | AB007177 <sup>d</sup>                        |
| RPL26  | 17              | 1     | cggtgattgtaccttctaaaacg     | tccaaaacaaataaacogcaat      | 344                 | RP_L26    | L07287                                       |
| RPL27  | 17              | 3     | CCCTGGACAAACTGTGTGTC        | CCACTTGTCTTGTGCTGCT         | 207                 | RP_L27    | L19527                                       |
| RPL27A | 11              | 3     | GTTAAACCTTGACAAATGTGG       | CCCCCAACACTCTTAACT          | 1000 <sup>c</sup>   |           | U14968                                       |
|        |                 |       | GTTAAACCTTGACAAATGTGG       | taggcactgctacggatagg        | 267                 | RP_L27A_1 | AB007178 <sup>d</sup>                        |
| RPL28  | 19              | 3     | ACCTCCTATGTGGGACCAC         | CTCTTCCACATCACAGGCTT        | 356                 |           | U14969                                       |
|        |                 |       | ACCTCCTATGTGGGACCAC         | atccctctccagcccatatt        | 198                 | RP_L28_1  | AB007179 <sup>d</sup>                        |
| RPL29  | 3 <sup>v</sup>  | 1     | GCCAAAGTCCAGAACCA           | tgtgggagaaaaaaggga          | 362                 | RP_L29    | U10248                                       |
| RPL30  | 8 <sup>a</sup>  | 2     | GGCCGCAAGAGACAGgt           | GTAGCGTTTCCGCACTTTTct       | 280 <sup>c</sup>    | RP_L30    | DL4521,L05095                                |
| RPL31  | 2               | 3     | GAATGTGCCATACCGAATCC        | TTAGTTTTCATCCACATGACTG      | 424                 |           | X15940                                       |
|        |                 |       | agttctccatcccataaagcc       | cctttttagacgggtctgtg        | 212                 | RP_L31_1  | AB007180 <sup>d</sup>                        |
| RPL32  | 3               | 2     | AGACCGATATGTCAAATTAAGgt     | GTTCGCGCAGTTACGct           | 650 <sup>c</sup>    | RP_L32    | K02060,X03342                                |
| RPL34  | 4               | 3     | CGTTTGACATACGACGTAG         | CACACACACATGCAGATTT         | 217                 |           | L38941                                       |
|        |                 |       | caaaatgctgaactactgactg      | CACACACACATGCAGATTT         | 116                 | RP_L34_1  | AB007181 <sup>d</sup>                        |
| RPL35A | 3 <sup>w</sup>  | 3     | CGGCAACCAACCAAAAC           | TCCTTGAGGGGTACAGCA          | 1800 <sup>c</sup>   |           | X52966                                       |
|        |                 |       | aagtgtatgggattacaggtgtg     | TCCTTGAGGGGTACAGCA          | 246                 | RP_L35A   | AB007182 <sup>d</sup>                        |
| RPL36A | X <sup>a</sup>  | 1     | agagcaaccatctctgtcc         | TGCTTGCCACACTTCTTACA        | 353                 | RP_L36A_1 | L35265                                       |
| RPL37  | 5               | 3     | CGAAGCTGTGCAAAATGTG         | TTAGATGAATGGATGCTGC         | 1800 <sup>c</sup>   |           | L11567                                       |
|        |                 |       | CGGAGATTCAgtaactgtt         | TTAGATGAATGGATGCTGC         | 1500 <sup>c</sup>   | RP_L37_1  | AB007183 <sup>d</sup> ,AB007184 <sup>d</sup> |
| RPL37A | 2               | 2     | GGTGCTTGGAGGTACAAgt         | GTCACAGCGGAAGTGGTact        | 1200 <sup>c</sup>   | RP_L37A   | DL4167,X66699                                |
| RPL38  | 17              | 3     | AGTTTAAAGTTGATGACACA        | TCATTTTCAGTTCTTCACTGC       | 611                 | RP_L38_1  | Z26876                                       |
|        |                 |       | togaggttggtgctagtactg       | TCATTTTCAGTTCTTCACTGC       | 281                 |           | AB007185 <sup>d</sup>                        |
| RPL40  | 19 <sup>y</sup> | 1     | gtgtcagttctcagactcccc       | agaaggagggcgactgag          | 220                 | RP_L40    | X56997                                       |
| RPL41  | 12              | 3     | ACTTGGCCTTCTCTGG            | CTCTTTTGGGCTTCAAG           | 879                 |           | Z12962                                       |
|        |                 |       | cagccataacctctgtaacta       | ggaacttccctctctctgtg        | 380                 | RP_L41_1  | AB007186 <sup>d</sup>                        |
| RPP0   | 12              | 3     | ATCATCAACGGGTACAAAG         | ATATCTCTGTGCGACTCTC         | 601                 |           | M17885                                       |
|        |                 |       | atcccaacaggaccacagt         | agcctttctgtcagaagca         | 271                 | RP_P0_1   | AB007187 <sup>d</sup>                        |
| RPP1   | 15              | 3     | TGGCAACGTCACATTTG           | GTCATCATCAGACTCTGGG         | 310                 | RP_P1     | M17886                                       |
| RPP2   | 11              | 2     | GACGTCAATGCCAGGgt           | GCAAGCTTGCCAAATACct         | 850 <sup>c</sup>    | RP_P2     | X55153,M17887                                |

<sup>a</sup>Upper- and lowercase letters denote exon and intron sequences, respectively.<sup>b</sup>Sequence from which STS was generated.<sup>c</sup>Size (in bp) estimated by agarose gel electrophoresis.<sup>d</sup>Sequence determined in this study.<sup>e</sup>Previously reported chromosomal assignments were confirmed: <sup>e</sup>Polakiewicz et al. 1995; <sup>f</sup>Nolte et al. 1996; <sup>g</sup>Fisher et al. 1990; <sup>h</sup>Antoine and Fried 1992; <sup>i</sup>Annali et al. 1995; <sup>j</sup>Davies and Fried 1993; <sup>k</sup>Feo et al. 1992; <sup>l</sup>Rhoads et al. 1986; <sup>m</sup>Jones et al. 1997; <sup>n</sup>Imai et al. 1994; <sup>o</sup>Kas et al. 1993; <sup>p</sup>Yon et al. 1993; <sup>q</sup>Mazuruk et al. 1996; <sup>r</sup>Van den Ouweland et al. 1992; <sup>s</sup>Adams et al. 1992; <sup>t</sup>Davies and Fried 1995; <sup>u</sup>Fan et al. 1997; <sup>v</sup>Garcia-Barcelo et al. 1997; <sup>w</sup>Colombo et al. 1996; <sup>x</sup>Oeltjen et al. 1995; <sup>y</sup>Webb et al. 1994.<sup>z</sup>Previously reported chromosomal assignment (Feo et al. 1992) was corrected.

## A MAP OF 75 HUMAN RIBOSOMAL PROTEIN GENES

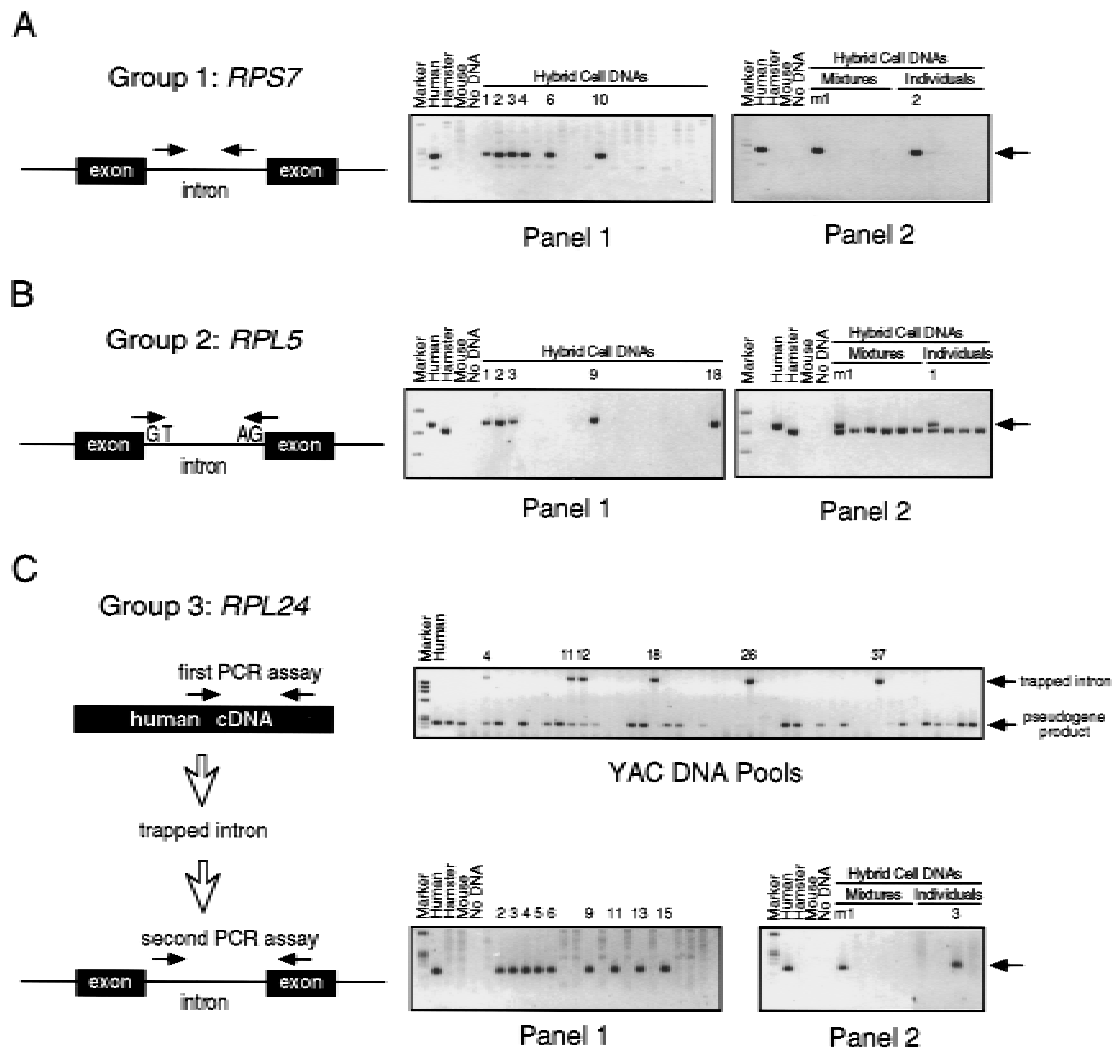


Figure 1 Intron trapping: derivation of STSs for and chromosomal assignment of representative group 1, group 2, and group 3 rp genes. (A) (Group 1) PCR primers were chosen from a previously sequenced intron of human *RPS7*. (Right) Results of testing of human/rodent somatic cell hybrid DNAs (NIGMS panels 1 and 2) for the *RPS7* STS; agarose gel stained with ethidium bromide. On each gel, the left-most lane contains size markers, and the next four lanes show results of PCR controls with human, hamster, or mouse genomic DNA, or no added DNA, as template. (Panel 1) 18 hybrid lines, each retaining multiple human chromosomes; 6 hybrids that tested positive for *RPS7* are numbered; the results map *RPS7* to chromosome 2. (Panel 2) 24 hybrid lines, most retaining a single human chromosome; we initially tested these hybrids in pools of four (here mixture m1 is positive) and then tested individual hybrids from the positive pool (the chromosome 2 hybrid is positive). (B) (Group 2) PCR primers were chosen from human *RPL5* cDNA sequence at a predicted splice site, with the dinucleotide GT appended to the 3' end of the forward primer and the dinucleotide CT (the complement of AG) appended to the 3' end of the reverse primer. (Right) Results of hybrid mapping, which assigned *RPL5* to chromosome 1. The arrowhead (extreme right) indicates the size of the human PCR product; a smaller, hamster product is also present in many lanes. (C) (Group 3) Mapping of *RPL24* involved two quite different PCR assays. The first PCR assay (top), with 45 YAC DNA pools as template, was designed to trap an intron with primers chosen from human cDNA sequence according to rules discussed in the text. Six YAC pools that yielded the higher molecular weight, trapped-intron product are numbered; many more pools yielded the lower molecular weight, pseudogene product. The control reaction with human genomic DNA as template yielded only the pseudogene product. Sequencing of the trapped intron made possible a second PCR assay specific to the functional *RPL24* gene (bottom); with the second assay, the gene was mapped to chromosome 3.

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ping strategy. A survey of all rp genes whose intron/exon structures had been determined in any two vertebrates revealed universal conservation of intron location but little conservation of intron sequence (analysis not shown; for examples, see Mameda et al. 1993; Annilo et al. 1995; Davies and Fried 1995). Thus, we could predict the sites of introns within human group 2 cDNA sequences by comparison with more thoroughly studied vertebrate homologs. These predictions enabled us to avoid coamplification of pseudogenes by choosing PCR primers that were likely to contain splice sites and to extend slightly into introns. We exploited the fact that introns usually have a GT dinucleotide at their 5' end and an AG dinucleotide at their 3' end. As diagrammed in Figure 1B, the forward and reverse primers were based on human cDNA sequence immediately preceding and following a predicted splice site, but with the dinucleotide GT appended to the 3' end of the forward primer and the dinucleotide CT appended to the 3' end of the reverse primer. Thus, both the forward and reverse primers extended two nucleotides into the intron. The resulting PCR product was a trapped intron.

We avoided splice sites where the human cDNA sequence is AG/GT, as in such cases processed pseudogenes would be expected to be perfect templates. In a few cases where primers immediately flanking one splice site had markedly different melting temperatures or were otherwise predicted to be incompatible, primers were selected from two consecutive splice sites (i.e., primers predicted to span the outer borders of two consecutive introns).

For group 3 genes, we had no prior information about splice sites in any vertebrate. These genes could not be mapped by use of the group 1 or group 2 methods. However, if we could trap rp gene introns using human cDNA sequence as our only starting information, we might then derive intron-based sequence tags that would identify the functional genes as distinct from their pseudogenes. We arrived at the scheme diagrammed in Figure 1C. Using a forward primer likely to derive from the penultimate exon and a reverse primer likely to derive from the final exon, we attempted to PCR amplify the final intron of each group 3 gene. The details of the strategy emerged from several considerations. We would have to confront not only cross-amplification of pseudogenes but also the possible failure of the functional gene to amplify because the sequence complementary to one (or both) of the PCR primers was interrupted by an intron. We noted that, for most human rp genes whose intron/exon structure has been determined, the 50 nucleo-

tides upstream of the termination codon are not interrupted by an intron. Thus, we chose the reverse primer from within a 50-bp region immediately upstream of the termination codon; this reverse primer was likely to derive from the final exon. We also noted that the terminal exon of human rp genes rarely extends >150 bp upstream of the termination codon, and that internal exons average 100 bp in length. Thus, we chose the forward primer from within the region 150–250 bp upstream of the termination codon; this forward primer was likely to derive from the penultimate exon or the one preceding. Further, we noted that about half of all splice sites in human ribosomal protein coding sequences are preceded by the dinucleotide AG. To reduce the probability that primers would span splice sites, we chose forward primers that did not include the sequence AG.

The rules just outlined were intended to (1) maximize the probability that PCR primer pairs would flank one or two introns (products containing three or more introns might be more difficult to amplify) and (2) minimize the probability that either primer would span a splice site. However, these rules for primer selection would not prevent amplification of closely related pseudogenes. We were concerned that during PCR amplification of human or human–rodent hybrid genomic DNA, competition between functional gene and pseudogene templates would occur; such competition would usually favor the pseudogenes because of their greater number, their high sequence similarity to the functional genes, and the smaller size of the resulting PCR product. In an effort to circumvent this problem, we tested PCR primer pairs on 45 pools of random human YACs, each pool comprising a random fifth of the human genome (Chumakov et al. 1995). (The YAC inserts averaged 0.6–1.0 Mb, and each pool contained 768 YACs.) We reasoned that among the 45 YAC pools, 9 or 10 should contain the functional gene and that, of these, one or more pools might be relatively free of competing pseudogenes. Such YAC pool roulette often yielded a jackpot (a clearly discernible, higher molecular weight PCR product, presumably containing one or more introns) in one or more YAC pools, even when use of the same primers on human genomic DNA yielded only pseudogene products (e.g., *RPL24*, as shown in Fig. 1C).

We then sequenced the higher molecular weight products obtained by PCR on YAC pools to confirm the presence of a genuine rp gene intron flanked by consensus splice sites. Indeed, by sequencing such PCR products, we identified a total of 55 new splice sites within 44 of the group 3 genes



(Table 2). For 34 of these genes, the PCR product contained a single intron, in which case we identified a single splice site. For the other 10 genes, the PCR product contained two (or, in one case, three) introns, in which case we identified two (or three) splice sites. All splice junction sequences conformed to the GT/AG rule and approximated the larger consensus sequence described by Mount (1982). The trapped introns varied in length from 75 to ~2100 bp. With the sequence of the trapped introns in hand, we were then able to design new PCR assays that amplified the functional rp genes but not their processed pseudogenes. These new, intron-based STSs were employed in subsequent mapping experiments.

### Chromosomal Assignments

In all, we succeeded in identifying PCR-assayable STSs for 75 rp genes, including all 19 group 1 genes, all 12 group 2 genes, and 44 of 49 group 3 genes. To assign each of these 75 rp genes to an individual human chromosome, we tested two panels of human–rodent hybrid cell line DNAs (Drwinga et al. 1993) for the presence of the corresponding STS. For each of the rp gene STSs the chromosomal assignments derived by use of the first and second hybrid panels were concordant. (Three examples are shown in Fig. 1.) In this manner, each of the 75 rp genes was unambiguously mapped to a single human chromosome (Table 1). Chromosomal assignments had been reported previously for 24 of these 75 genes, and in 23 cases we confirmed these prior studies. In only one case (*RPS17*, on chromosome 15) do our results contradict a previous assignment.

### Fine Localization

We employed two methods, radiation hybrid (RH) mapping and YAC/STS content analysis, to localize more precisely the 75 chromosomally assigned rp genes. For RH mapping, we scored for the presence or absence of each of the rp gene STSs in 91 human–hamster hybrid cell lines comprising the GeneBridge 4 whole-genome RH panel (Walter et al. 1994). This RH panel had been used previously to construct a comprehensive, STS-based map of the human genome (Hudson et al. 1995). Analysis of the rp gene STS typing results allowed us to position 73 of the rp genes on this pre-existing map (Fig. 2).

In parallel, we attempted to place the chromosomally assigned rp genes on a pre-existing YAC/STS content map of the human genome (Hudson et al.

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1995). In this case, the CEPH YAC library (Chumakov et al. 1995) was screened by PCR to identify clones containing human rp genes. We identified a total of 222 YACs that carry 55 different rp genes. Information as to the identities, STS content, and chromosomal location of these 222 ribosomal protein-encoding YACs is available at an Internet site maintained by the Whitehead Institute/MIT Center for Genome Research ([http://www-genome.wi.mit.edu/cgi-bin/contig/phys\\_map](http://www-genome.wi.mit.edu/cgi-bin/contig/phys_map)).

By integrating the results of our RH and YAC mapping efforts, we were able to place all 75 chromosomally assigned rp genes on the map of the human genome (Fig. 2). Regional assignments had been reported previously for some of these genes; in only one case (*RPL29*, on chromosome 3p) do our results conflict with a previous regional localization.

### DISCUSSION

The ribosome is the most complex mammalian structure to have been completely described at the level of nucleotide and amino acid sequence (Wool 1996; Maidak et al. 1997). We now know the chromosomal locations, in humans, of the genes encoding all 4 RNAs and 75 of an estimated 80 proteins comprising this elaborate, protein synthesis machine (Fig. 2). Most mammalian ribosomal proteins have recognized homologs in prokaryotes, where rp genes are organized into a small number of operons, with as many as 11 ribosomal proteins under the control of a single promoter (Nomura et al. 1984). In contrast, there is little evidence of rp gene clustering in mammals (Feo et al. 1992)—a conclusion that our results confirm and extend. Both human sex chromosomes and at least 20 autosomes (all but chromosomes 7 and 21) carry one or more rp genes (Table 1; Fig. 2). Only the presence of 12 rp genes on chromosome 19, which constitutes only two percent of the haploid genome (Morton 1991), is notably at odds with a random distribution of rp genes throughout the human genome. Chromosome 19 is known to have a high gene density (Schuler et al. 1996), and even here, the 12 rp genes are scattered. With 55 rp genes mapped to YAC clones, we found only two examples of multiple rp genes residing on the same YAC clone: *RPS26* and *RPL41* (on chromosome 12), and *RPS11* and *RPL13A* (on chromosome 19). If one considers both ribosomal RNAs and proteins, it is apparent that virtually every human chromosome (except perhaps chromosome 7) contributes one or more components to the ribosome (Fig. 2). Though encoded at dispersed genomic sites, the ribosome's myriad components are apparently as-

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sembled with stoichiometric precision. Regulated coproduction of the components could, in theory, be achieved in several ways. The clustering of *rp* genes in operons, as in bacteria (Nomura et al. 1984), is evidently not an important means of regulated coproduction in humans. *Trans*-acting regulatory mechanisms, both transcriptinal and translational, have been argued to play a substantial role in coordinating production of ribosomal components in mammals (Hariharan et al. 1989; Meyuhas et al. 1996), though feedback mechanisms, if any, remain to be elucidated. Alternatively, some ribosomal components may simply be produced in excess, with molecules not incorporated into ribosomes being discarded.

A few human *rp* genes remain to be mapped. In five cases (*RPL14*, *RPL22*, *RPL35*, *RPL36*, and *RPL39*), we were unable to trap verifiable introns despite repeated efforts; we did not map these five genes. We anticipate that these genes will be mapped as more information about their gene structures becomes available. [One of these genes, *RPL22*, has been assigned to chromosome 3 (Nucifora et al. 1993).] As a rule, each mammalian ribosomal protein is encoded by a single functional gene (Dudov and Perry 1984; Wagner and Perry 1985; Kuzumaki et al. 1987), but we cannot exclude the possibility that a second functional gene may exist in some cases. Indeed, functionally interchangeable isoforms of *RPS4* are encoded by the human X and Y chromosomes (Fisher et al. 1990; Watanabe et al. 1993). In the case of *RPL36A*, a functional, intron-bearing gene is located on the X chromosome (Oeltjen et al. 1995), but analysis of cDNA sequences suggests that a second functional gene may exist elsewhere in the genome, as yet unmapped (N. Kenmochi et al., unpubl.).

## Implications for the Human Genome Project

The methods we employed for STS generation via intron trapping should be of general use in mapping genes with processed homologs. Recently, PCR-

Table 2. Newly Identified Splice Sites in Human Ribosomal Protein Genes

| Gene   | Exon                          | Intron size (bp)  | Exon                   |
|--------|-------------------------------|-------------------|------------------------|
| RPSA   | 830 <sup>a</sup><br>TTCCTACTG | 212               | ctcttaacag AAGACTGGAG  |
|        | 733<br>gtatgtatca             |                   |                        |
| RPS2   | 733<br>GGCAACTTCG             | 226               | tggttttcag CCAAGGCCAC  |
|        | 690<br>gtaggtgggc             |                   |                        |
| RPS3A  | 690<br>AAGTTTGAAT             | 187               | tgccttttag TGGGAAGCT   |
|        | 484<br>gtaaagtgaga            |                   |                        |
| RPS5   | 484<br>TGTCGAACAG             | 1100 <sup>b</sup> | gtcttcctag GGCATCTGGC  |
|        | 583<br>gtgagcctgg             |                   |                        |
| RPS9   | 583<br>TGCTGCCAAG             | 86                | ctccttcag GGCCTCTGA    |
|        | 132<br>gtgggtgagg             |                   |                        |
| RPS10  | 132<br>AAGCTGATCG             | 205               | tccccaccag GCGATATGG   |
|        | 166<br>gtgagtgccc             |                   |                        |
| RPS11  | 166<br>GGCCATGCAG             | 232               | tgccttacag TCCTCAAGT   |
|        | 368<br>gtaggagggt             |                   |                        |
| RPS12  | 368<br>CCTGCTTCAG             | 1464              | gcctccacag GGCCTCCAG   |
|        | 313<br>gtgagcgcag             |                   |                        |
| RPS13  | 313<br>CCTAATTAAG             | 400 <sup>b</sup>  | ctcccaatag GTTGAATGACA |
|        | 415<br>gtaagctgc              |                   |                        |
| RPS15A | 415<br>AGTAGTTAAG             | 500 <sup>b</sup>  | tttttttaag GACTATGGCA  |
|        | 454<br>gtaagtcacc             |                   |                        |
| RPS19  | 454<br>ATTGGAATA              | 629               | gcttttcag TGAATCATCT   |
|        | 133<br>gtatgtatca             |                   |                        |
| RPS20  | 133<br>ATGGAAGCATG            | 750 <sup>b</sup>  | cttttggtag GTTACATTGG  |
|        | 378<br>gtatgtctgc             |                   |                        |
| RPS21  | 378<br>ACCAAGATGG             | 480 <sup>b</sup>  | ctctccacag CGGCCCAAA   |
|        | 290<br>gtaacgagg              |                   |                        |
| RPS23  | 290<br>GCCTACCAAG             | 423               | ttgtttacag ACTTTGAGAA  |
|        | 152<br>gtaaagtaaa             |                   |                        |
| RPS26  | 152<br>CGTGGCCGAG             | 168               | ctcttttcag GTTGCAAGG   |
|        | 224<br>gtgagctggg             |                   |                        |
| RPS27  | 224<br>TGCTAGGATG             | 392               | ttttcttaag GGTCAGTCAG  |
|        | 298<br>gtgagtggtt             |                   |                        |
| RPS28  | 298<br>CTTTAATTCAG            | 145               | tttattccag GAAATGATG   |
|        | 204<br>gtgagtatct             |                   |                        |
| RPS29  | 204<br>GTCTTCGATG             | 700 <sup>b</sup>  | tattccttag CCTATGTCCT  |
|        | 135<br>gtaaagtggt             |                   |                        |
| RPS30  | 135<br>AAATGCCAG              | 344               | tttctttcag GATGCTATAA  |
|        | 346<br>gtgaggagac             |                   |                        |
| RPS31  | 346<br>ATATATATAG             | 465               | tgcttttcag GTGGATGAGA  |
|        | 241<br>gtgagccagt             |                   |                        |
| RPS32  | 241<br>CTGCTCCCTG             | 180               | ctgttttcag GGCTTCGAT   |
|        | 192<br>gtgggtgcaa             |                   |                        |
| RPS33  | 192<br>TTTCATTAAG             | 2100 <sup>b</sup> | tttttttcag TTGGACTAAA  |
|        | 1053<br>gtagcgtct             |                   |                        |
| RPL3   | 1053<br>CCTCCGCAAG            | 323               | tggcccatag TCCTTCCTGG  |
|        | 1173<br>gtgaggcagg            |                   |                        |
| RPL6   | 1173<br>AGCATTCATG            | 206               | ctccccacag GGACACTGA   |
|        | 735<br>gtgagcacct             |                   |                        |
| RPL8   | 735<br>AGAAAAAGAG             | 476               | ctttatttag AAATATGAGA  |
|        | 181<br>gtaagtttct             |                   |                        |
| RPL9   | 181<br>CATCGTCAAG             | 77                | ctgcccacag GACATCATCC  |
|        | 323<br>gtgcccgaag             |                   |                        |
| RPL10  | 323<br>GGCAAGGAGG             | 277               | ccccttcag CCCAGTCAA    |
|        | 478<br>gttagcatog             |                   |                        |
| RPL11  | 478<br>TCAAATTCAG             | 214               | actcctatag CGCCTTCAT   |
|        | 595<br>gtttgtatgt             |                   |                        |
| RPL12  | 595<br>GATCTAAGAG             | 307               | ttttttacag TTACCTGGCT  |
|        | gtaagttcct                    |                   |                        |

(Continued on facing page.)

based typing of RH panels (or YAC libraries) was used to map many thousands of gene-based STSs (Schuler et al. 1996). These high-volume gene mapping efforts relied on STSs drawn from 3' untranslated regions of genes (Berry et al. 1995). This method minimized the chance that PCR products



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Table 2. (Continued)

| Gene   | Exon                         | Intron size (bp)      | Exon                       |
|--------|------------------------------|-----------------------|----------------------------|
| RPL10  | 506<br>COGOCAGAG gtagtagtg   | - - 75                | - - tgctccttag ATOCACATCT  |
| RPL11  | 160<br>TTTTCCAAAG gtgagtagtc | - - 850 <sup>b</sup>  | - - ttccctgcag CTAGATACAC  |
| RPL12  | 447<br>GAAGTCTCTG gtaagagcag | - - 347               | - - ttacttaaag GAACCATTA   |
| RPL13  | 528<br>CGTCCGGAAC gtaagtgaac | - - 492               | - - tctcttctag GTCATATAA   |
| RPL15  | 188<br>GOCAGCAAG gtaagtgtatc | - - 380 <sup>b</sup>  | - - ttgtgtgtag GTTACGTTAT  |
| RPL17  | 645<br>GACAAAAAG gtaataagt   | - - 750 <sup>b</sup>  | - - ctttccccag ATATCCACAG  |
| RPL18A | 347<br>AOCAGTCT gtaagtgtcc   | - - 750 <sup>b</sup>  | - - ctttcaaaag AOCAGACAT   |
|        | 457<br>GCAGTTCCAC gtgagtgtcc | - - 143               | - - ttccctcag GACTCCAGA    |
| RPL21  | 422<br>AAAGCGCCAG gtaagaaatt | - - 101               | - - cctttaatag CCGTCTCCAC  |
| RPL23  | 365<br>GAGATGAAAG gtaggaatc  | - - 147               | - - tttaatgcag GTTCTGCCAT  |
| RPL24  | 368<br>AAGCTATCAG gtgaggaatg | - - 278               | - - taccocacag GGCTGCTAAG  |
|        | 432<br>TGCTGCTAAG gtaattatgg | - - 1100 <sup>b</sup> | - - cttttttacag GCACCTACAA |
| RPL27  | 379<br>TTGAGAGAG gtaagttagt  | - - 88                | - - cccctcctag ATACAGACA   |
| RPL27A | 334<br>GGTGGATCG gtaagttaat  | - - 800 <sup>b</sup>  | - - gttcttctag GGCTACTACA  |
| RPL28  | 351<br>CCTGGCATG gtgagctggg  | - - 195               | - - ccgccccag GCAGCCATCC   |
| RPL31  | 354<br>CTTTCAAAA gtaagttctc  | - - 279               | - - tctttttacag TCTACAGACA |
| RPL34  | 85<br>AACTAGGCT gtaagtattt   | - - 80                | - - cacttttctag GTCCGAAAC  |
| RPL35A | 371<br>AATCCGAGTG gtgagtatgg | - - 1700 <sup>b</sup> | - - ttccctgcag ATGCTGTACC  |
| RPL37  | 145<br>AAGAGAAAGT gtaagtaaca | - - 205               | - - tatgttttag ATAACTGGAG  |
|        | 230<br>GCAGATTCAG gtacagtttg | - - 1400 <sup>b</sup> | - - catcctgcag GCATGGATTC  |
| RPL38  | 297<br>CTGCCCCCG gtgagtggc   | - - 495               | - - ttccctctag GTTTGGCAGT  |
| RPL41  | 70<br>ATTTTTTTGG gtgagtgttt  | - - 116               | - - ttccctgtag AAACCTCTGC  |
|        | 95<br>GAGGCCAAG gtgagcggtt   | - - 388               | - - oggttgctag TGGAGGAAGA  |
|        | 118<br>GAATGCCAG gtaagtgtgag | - - 271               | - - tgcttttctag GCTGAGGCG  |
| RPF0   | 869<br>TGCTGAAAAG gtaaaaggat | - - 389               | - - ctcccttctag GTCAAGGCGT |
| RPF1   | 394<br>GCTGCTCCAG gtaggaaca  | - - 140               | - - ttttttctag CTGAGGAGA   |

Upper- and lowercase letters denote exon and intron sequences, respectively.

For DDBJ/EMBL/GenBank accession nos., see Table 1.

<sup>a</sup>Last nucleotide in exon is numbered according to position in cDNA sequence.

<sup>b</sup>Size estimated by agarose gel electrophoresis.

would contain introns, thereby reducing the size of PCR products and increasing amplification efficiency. In contrast, our rp gene STSs were designed to span or lie within introns, to identify functional, intron-bearing genes as distinct from processed pseudogenes. Although the procedure for deriving STSs from 3'-untranslated regions of genes is simpler—a requirement of high-volume genomic studies—the procedure would likely yield erroneous

mapping results when applied to genes with abundant processed pseudogenes. Had we applied this strategy to rp genes, we would have completely overlooked the functional genes, which are less efficient than their intron-less pseudogene derivatives as templates for PCR amplification with cDNA-based primers. Although an error rate of only 1% has been claimed for the high-volume, STS-based gene mapping efforts (Schuler et al. 1996), these error estimates took no account of processed pseudogenes, the impact of which could be significant if a sizable fraction of all genes give rise to processed pseudogenes. The methods we employed for STS generation via intron trapping should be of use in efforts to systematically map genes with a propensity to generate processed pseudogenes, that is, housekeeping and other genes that are abundantly expressed in the germ line.

#### Ribosomal Protein Defects in Human Disease

Evolutionary and genetic considerations lead us to predict roles for rp genes in human disease. Ribosomal proteins are highly conserved among eukaryotes and prokaryotes. Virtually all mammalian ribosomal proteins have counterparts (with 40%–88% amino acid identity) in the yeast ribosome (Wool et al. 1996). Of the 78 rat ribosomal proteins whose amino acid sequence is known, at least 49 have recognizable homologs in the archaeobacterial ribosome (Wool et al. 1996). Among multicellular animals, the consequences of mutations in rp genes have been explored most thoroughly in *Drosophila*. Here, mutations resulting in reduced expression of individual ribosomal

proteins yield the *Minute* phenotype. Because a full complement of ribosomal proteins is required to assemble a functional, stable ribosome, *Minute* cells probably contain fewer ribosomes and thus have less capacity for protein synthesis (Kay and Jacobs-Lorena 1987). Conservation of ribosomal proteins among eukaryotes, combined with sequence studies, indicate that *Drosophila* and human ribosomes are extremely similar. Thus, it is likely

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that quantitative deficiencies in human ribosomal proteins, as in *Drosophila*, will result in reduced translational capacity and thereby yield specific, reproducible phenotypes. If specific human pheno-

types do result from ribosomal protein deficiencies, those phenotypes may or may not resemble the *Drosophila Minute* phenotype.

The present mapping study was motivated by

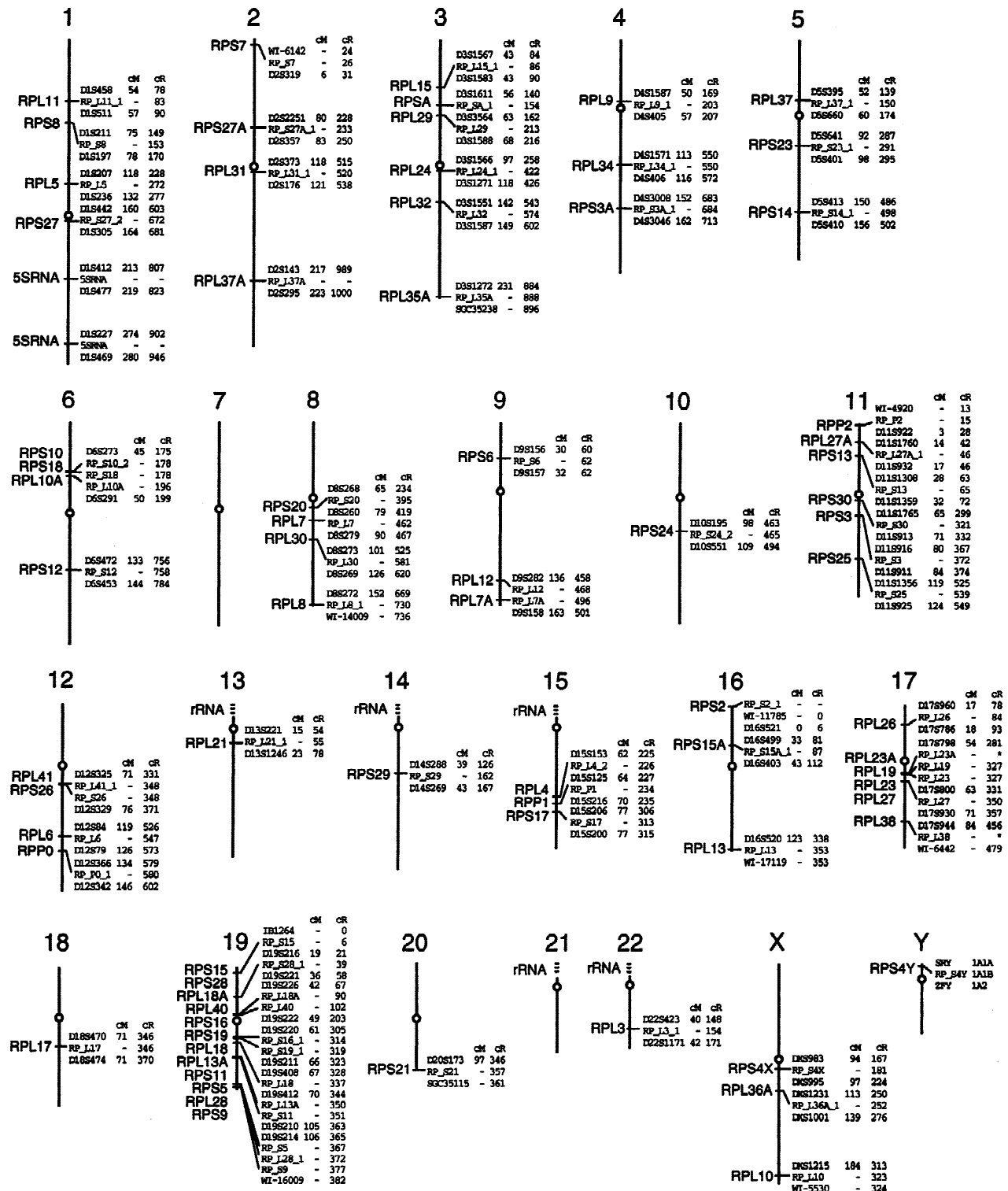


Figure 2 (See facing page for legend.)

## A MAP OF 75 HUMAN RIBOSOMAL PROTEIN GENES

the possibility that ribosomal protein mutations contribute to human disease, including Turner syndrome, other chromosomal birth defects, and Mendelian disorders. As yet, no human disorder has been traced definitively to a ribosomal protein mutation. Having a map of the rp genes will facilitate the search for mutations and roles in human disease, including monosomies and various Mendelian disorders.

*Monosomies*

Turner syndrome is a complex human phenotype associated with monosomy X (Ford et al. 1959). Embryos with a 45,X karyotype develop as females with poor viability in utero, and those that survive exhibit short stature, ovarian failure, and specific anatomic abnormalities that may include lymphedema, webbing of the neck, and cubitus valgus (Lippe 1991). These phenotypes likely result from the haploinsufficiency of certain genes that are common to the X and Y chromosomes and that escape X inactivation (Ferguson-Smith 1965); specific XY gene pairs probably account for particular components of this complex phenotype (Zinn et al. 1993). On the basis of studies of individuals carrying partial Y chromosomes, we and our colleagues have previously suggested that ribosomal protein S4, encoded by both the X and Y chromosomes, is an important factor in Turner syndrome (Fisher et al. 1990). Specifically, our results suggest that a quantitative deficiency of RPS4 may be responsible for the lymphedema and neck webbing observed in 45,X individuals, and perhaps also for the poor viability of 45,X fetuses (L. Brown, C. Raut, and D.C. Page, unpubl.). Following the principle, from *Drosophila*, that translational deficiencies stemming from any of a large number of rp genes yield a consistent, reproducible phenotype, we predict that deficiencies

of autosomal rp genes may yield Turner-like phenotypes in humans. Given the dispersion of rp genes throughout the human genome, virtually all monosomies and many partial monosomies will entail heterozygous deficiencies of one or more rp genes. We speculate that ribosomal protein deficiencies contribute to the abnormal development and poor viability of monosomic human fetuses and perhaps also to phenotypes observed in children born with partial monosomies. Thus, a search for correlations between ribosomal protein loci and haploinsufficient or haplolethal regions of the human genome should be initiated. [Note: The hypothesized role of RPS4 in Turner syndrome is controversial. Some investigators have concluded that RPS4 is not a Turner gene, because many Turner patients with structurally abnormal X chromosomes have two or more active copies of RPS4X (Just et al. 1992; Geerkens et al. 1996). However, as most such patients do not exhibit the particular Turner features that we attribute to RPS4 deficiency, these observations are consistent with our hypothesis. Omoe and Endo (1996) have suggested that RPS4 is perhaps not involved in Turner syndrome, because some mammalian species that have no RPS4Y gene on their Y chromosome exhibit an XO phenotype that is reminiscent of human Turner syndrome. However, because these XO animals do not exhibit the particular Turner features that we attribute to RPS4 deficiency, Omoe and Endo's observations are also consistent with our hypothesis.]

*Mendelian Haploinsufficiencies*

Quantitative deficiencies of individual ribosomal proteins could possibly result from either gross chromosomal deletions or point mutations in individual rp genes. In the latter case, resulting haploinsufficient traits might display simple Mendelian

**Figure 2** A map of genes encoding the human ribosome. The 22 autosomes and two sex chromosomes are shown as vertical lines, on which are positioned 75 rp genes (RP...), five ribosomal RNA gene clusters (rRNA), and two 5S RNA gene clusters (5SRNA). To the right of each chromosome are listed rp gene STSs, nearby markers, and approximate distances (in centiMorgans and/or centiRays) from the most distal short-arm marker on maps constructed at Génethon and the Whitehead Institute/MIT Center for Genome Research (Hudson et al. 1995). (Maps shown as of November 1997; for updated maps see <http://www-genome.wi.mit.edu/cgi-bin/contig/phys.map>). Because of the inherently statistical nature of RH mapping, we have high confidence in marker orders only where markers are separated by at least 15 centiRays. On distal 19q, for example, RPS5, RPL28, and RPS9 appear to be clustered within 10 centiRays and thus cannot be ordered with confidence. Our present data are most consistent with the order RPS5—RPL28—RPS9—qter (as shown), but higher resolution mapping experiments, while confirming the proximity of the three genes, strongly suggest the order RPS9—RPL28—RPS5—qter (N. Kenmochi, G. Lennon, S. Higa, and L. Ashworth, unpubl.). For the Y chromosome, where no genetic map is available, deletion map intervals (Vollrath et al. 1992) are listed. Our assignment of RPL29 to 3p conflicts with a recent report that it maps to 3q29-qter (Garcia-Barcelo et al. 1997). (\*) On chromosome 17, RPL23A and RPL38 were localized to the indicated intervals, but their distances from flanking markers could not be meaningfully estimated.

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transmission, probably appearing like autosomal dominant characters in human pedigrees. Because, in *Drosophila*, heterozygotes for loss-of-function mutations in any of a large number of rp genes display the *Minute* phenotype, one might expect the phenotype in human heterozygotes to be similarly consistent despite genetic heterogeneity. We predict that this autosomal dominant phenotype would encompass the components of the 45,X phenotype that are likely attributable to RPS4 deficiency, namely, lymphedema, neck webbing, and, perhaps, reduced fetal viability.

Particularly intriguing is the possibility that Noonan syndrome might be attributable to ribosomal protein deficiency. Many investigators have drawn attention to similarities between the Noonan and Turner phenotypes (Allanson 1987). As one would predict for a *Minute*-like ribosomal protein deficiency, Noonan syndrome is inherited in apparently autosomal dominant fashion, and it is genetically heterogeneous. We note that the *RPL6* and *RPPO* genes map to a region of chromosome 12 implicated, by linkage analysis, in a large pedigree with Noonan syndrome (Jamieson et al. 1994).

#### Other Mendelian Disorders

In considering the potential range of human disorders that might be caused by rp gene mutations, it is important to recognize that some ribosomal proteins may have additional, extraribosomal functions. Wool (1996) has suggested that, during evolution, proteins of diverse function were recruited to the ribosome to stabilize rRNA or otherwise enhance translation, and that, in some cases, these recruited proteins have also retained their ancestral functions. For example, in mammals and *Drosophila*, RPS3 functions as both a ribosomal protein and an endonuclease (Wilson et al. 1994; Kim et al. 1995; Yacoub et al. 1996). In *Drosophila*, RPS6 functions as a tumor suppressor in the hematopoietic system (Watson et al. 1992), and RPS2 functions in oogenesis (Cramton and Laski 1994), perhaps reflecting extraribosomal functions for these proteins. Given these precedents, human geneticists should not overlook the possibility that ribosomal protein mutations could directly perturb diverse cellular functions, without affecting translation, and thereby produce disease.

## METHODS

### DNA Sequences and Nomenclature

A complete catalog of rat ribosomal protein cDNA sequences

is available (Wool et al. 1996). We used these rat cDNA sequences to query the GenBank, EMBL, and DDBJ databases for human rp gene sequences (cDNA and genomic). The amino acid sequences of homologous rat and human ribosomal proteins are, on average, 99% identical. We refer to the human ribosomal proteins using the established rat nomenclature (Wool et al. 1996).

### PCR Primers and Conditions

For most rp genes, the primer selection rules described in the Results proved workable and effective. However, for a few group 3 genes, these rules were too stringent to permit selection of primer pairs, or the primers selected failed to amplify a higher molecular weight product with YAC pools as template. In several such cases, we were able to select satisfactory pairs by picking the forward primer from a larger target region (150–300 bp upstream of the termination codon) or by allowing the forward primer to contain a single AG dinucleotide. In all cases, PCR primer pairs were selected by use of the PRIMER program (S. Lincoln, M.J. Daly, E.S. Lander, Whitehead Institute); optimal oligonucleotide  $T_m$  was set at 58°C and the optimal primer length was set at 20 nucleotides.

PCR was performed in 20- $\mu$ l volumes containing 30–50 ng of template DNA, 10 pmole each of forward and reverse primers, 0.1 mM dNTPs, 10 mM Tris-Cl (pH 8.2), 50 mM KCl, 1.5 mM  $MgCl_2$ , 5.0 mM  $NH_4Cl$ . Reaction mixes were first heated at 90°C and, then, 1 unit of *Taq* DNA polymerase was added. Cycling conditions included an initial denaturation at 94°C for 3 min, followed by 35 cycles of 94°C for 1 min, 61°C for 1 min, 72°C for 1 min, and a final extension at 72°C for 5 min.

### Physical Mapping

RP gene STSs were chromosomally assigned by use of National Institute of General Medical Sciences (NIGMS) human-rodent hybrid cell line panels 1 and 2 (Drwina et al. 1993).

To place rp gene STSs on an existing RH map of the human genome (Hudson et al. 1995), we tested the hybrids of the GeneBridge 4 panel (Walter et al. 1994) in duplicate, by PCR, and analyzed the results using RHMAPPOR software (Hudson et al. 1995).

To place RP gene STSs on an existing YAC/STS content map of the human genome (Hudson et al. 1995), we screened 25,344 YACs (plates 709–972) from the CEPH library (Chumakov et al. 1995) using methods described previously (Hudson et al. 1995).

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